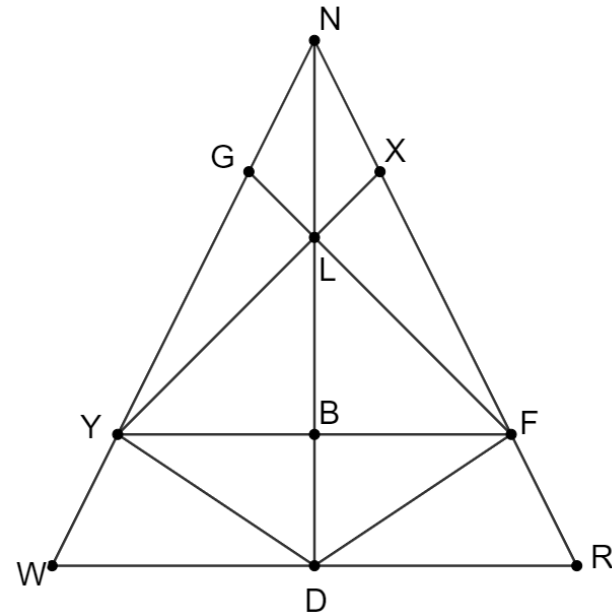


1. Triangle NRW is shown where $\overline{LB} \perp \overline{YF}$ and $\overline{DB} \perp \overline{WR}$.

Given: $\overline{LB} \perp \overline{YF}$, $\overline{DB} \perp \overline{WR}$

Prove: $\triangle YBN \sim \triangle WDN$



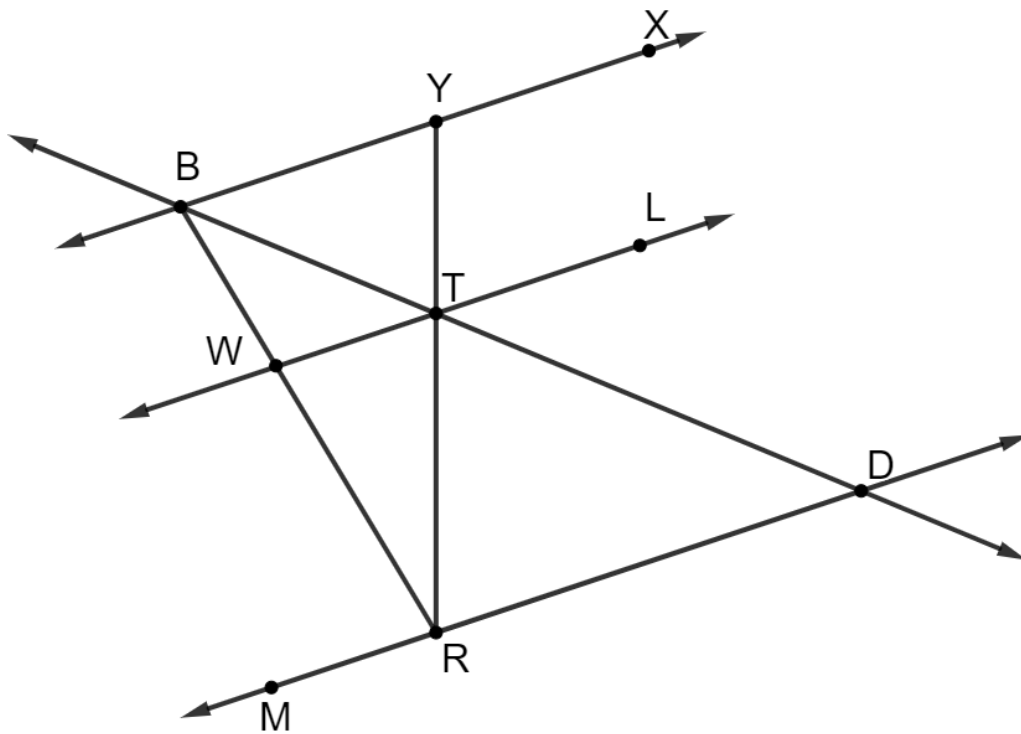
An incomplete proof is shown.

Statement	Reason
1. $\overline{LB} \perp \overline{YF}$, $\overline{DB} \perp \overline{WR}$	1. Given
2. $\angle YBN$ is a right angle; $\angle WDN$ is a right angle	2.
3. $m\angle YBN = 90^\circ$ and $m\angle WDN = 90^\circ$	3. Definition of a right angle
4. $m\angle YBN = m\angle WDN$	4. Transitive property of equality
5.	5. Reflexive Property
6. $\triangle YBN \sim \triangle WDN$	6.

What is the reason for step 6?

Reason for step 6	AA similarity
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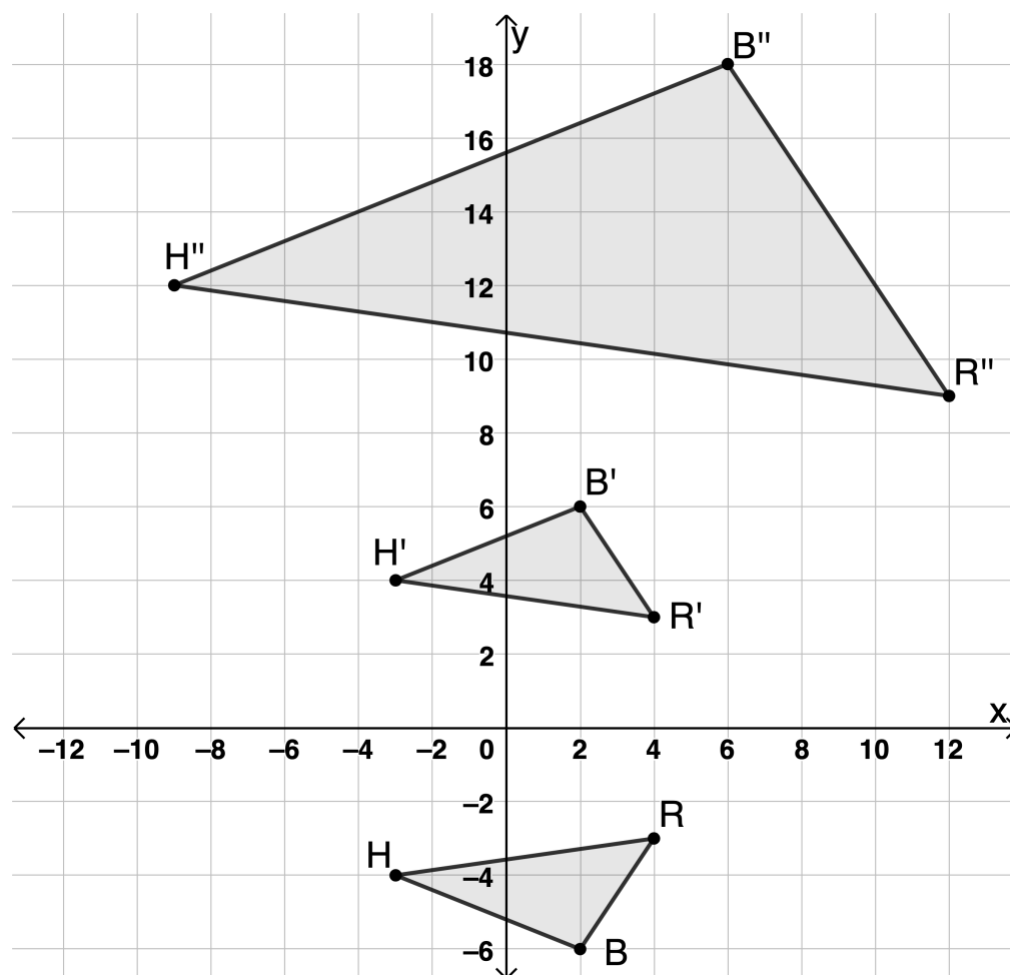
2. A diagram is shown.



For each set of given information, complete the missing information in the table.

	Given Information	Triangles	Are They Similar?	Theorem or Postulate Used
a.	$\frac{BT}{BD} = \frac{BW}{BR}$	$\triangle DBR$ and $\triangle TBW$	☐ Yes ☐ No	SAS similarity
b.	$\overleftrightarrow{BX} \parallel \overleftrightarrow{WL} \parallel \overleftrightarrow{MD}$ and $\frac{BW}{WR}$	$\triangle TWB$ and $\triangle TWR$	☐ Yes ☐ No	
c.	$\overleftrightarrow{BX} \parallel \overleftrightarrow{MD}$	$\triangle YTB$ and $\triangle RTD$	☐ Yes ☐ No	AA similarity
d.	$\frac{BR}{WR} = \frac{YR}{TR} = \frac{YB}{TW}$	$\triangle YRB$ and $\triangle TRW$	☐ Yes ☐ No	SSS similarity

3. Triangles HRB , $H'R'B'$, and $H''R''B''$ are shown on the coordinate grid.



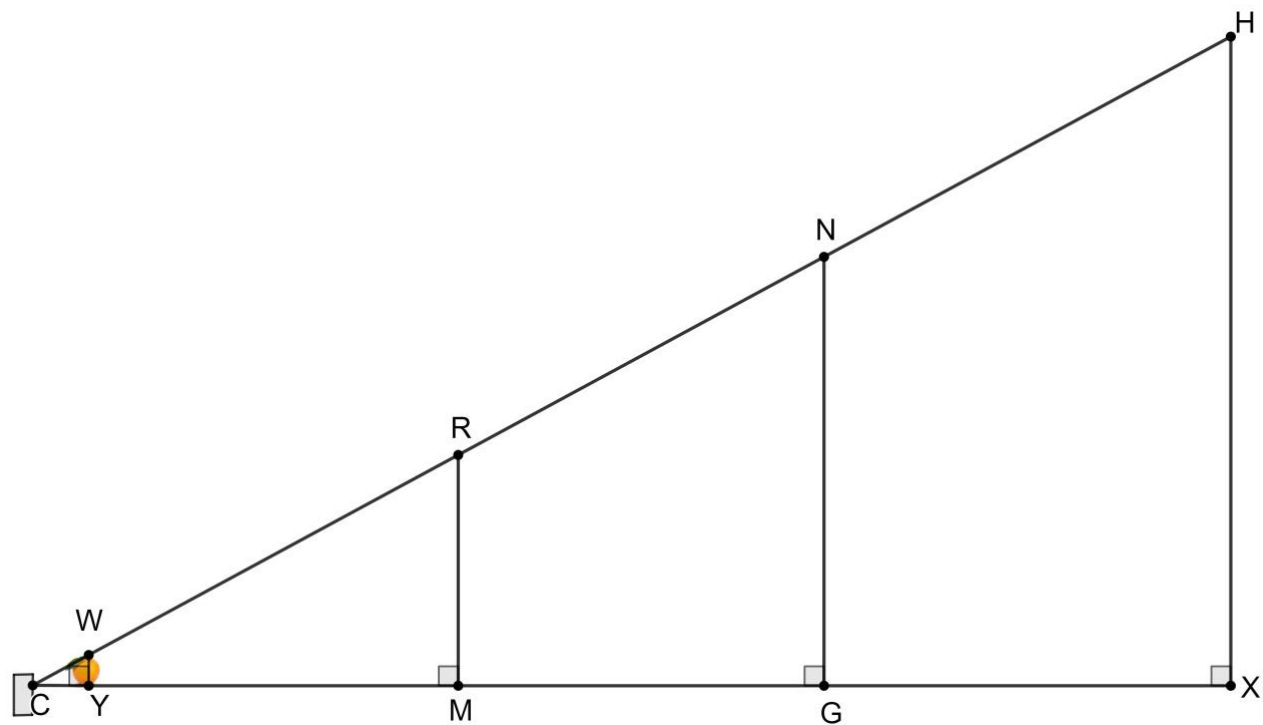
- a. Complete the table using algebraic descriptions so that the sequence of transformations will map $\triangle HRB$ onto $\triangle H''R''B''$.

Sequence	Algebraic Description
Transformation 1	$(x, y) \rightarrow (x, -y)$
Transformation 2	$(x, y) \rightarrow (3x, 3y)$

- b. Use the definition of similarity in terms of non-rigid transformations to justify that $\triangle HRB \sim \triangle H''R''B''$.

The sequence of transformations that map $\triangle HRB$ onto $\triangle H''R''B''$ include a reflection and a dilation. Because the sequence is a rigid motion and a dilation, the definition of similarity in terms of non-rigid transformations states that the triangles are similar.

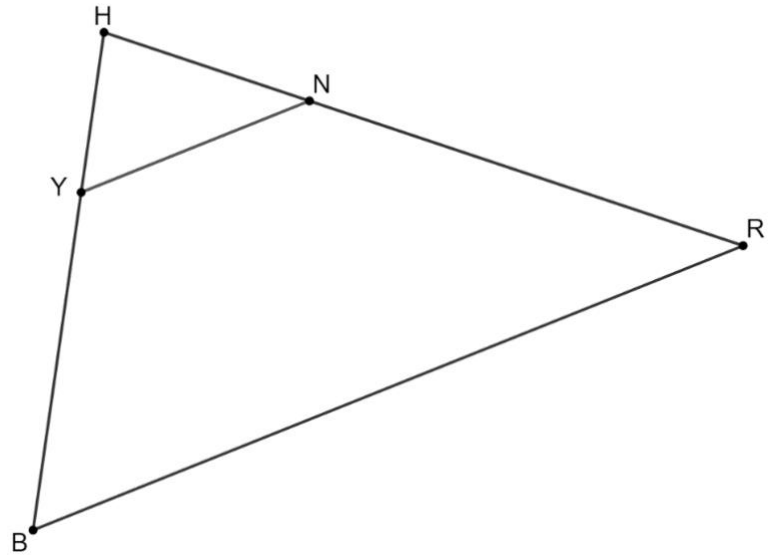
4. To create an image for her social media account, Shintasha places her phone close to an orange so that when the photo is taken the orange appears to be larger. Shintasha stands at various distances from the phone to determine the different heights the orange will appear compared to her. Her diagram is shown, where $WY = 6.3$ centimeters (cm), $CY = 10.5$ cm, $YM = 59.5$ cm, $MG = 52.5$ cm, and $GX = 73.5$ cm.



Determine the height the orange will appear when Shintasha stands at each of the points. Show your work by writing any equations you used to determine the height.

	Point	work	Height (cm)
a.	M	$\frac{6.3}{10.5} = \frac{x}{70}$	42
b.	G	$\frac{6.3}{10.5} = \frac{x}{122.5}$	73.5
c.	X	$\frac{6.3}{10.5} = \frac{x}{196}$	117.6

5. A paleontologist is setting up boundaries to dig for fossils. $\triangle HYN$ represents the area where invertebrate fossils have already been found. The paleontologist extends the dig area by drawing $\overline{BR}$ parallel to $\overline{YN}$ so that $\frac{YN}{BR} = 0.32$.



The paleontologist uses the diagram and the given information to extend the dig site. All measurements are in meters (m).

$HY = (6x + 4)$ m, $HN = (11x - 19.5)$ m, $NR = (18.5x)$ m, and $YN = 85$ m.

- a. Determine the distance, in meters, between points B and R . Show your work.

$$\frac{85}{BR} = 0.32 \quad BR = 265.625$$

- b. Determine the value of x . Show your work.

$$\frac{(11x-19.5)}{(29.5x-19.5)} = 0.32 \quad x = 8.5$$

- c. Determine the distance, in meters, between points B and Y . Show your work.

$$HY = 6(8.5) + 4 = 55$$

$$\frac{HY}{HB} = 0.32 \quad \frac{55}{HB} = 0.32 \quad HB = 171.875$$

$$BY = 171.875 - 55 = 116.875$$